

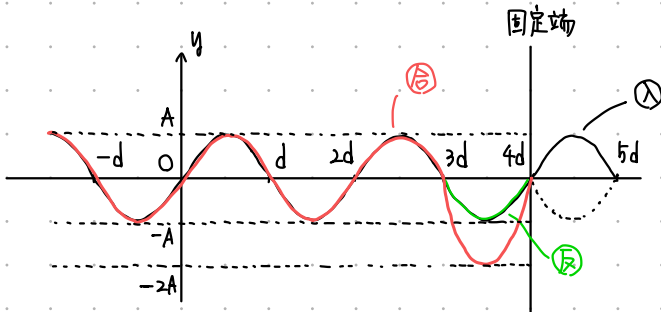
1

向1 $\lambda = 2d$

向2 $C = \frac{\lambda}{T} = \frac{2d}{T}$

向3 $t = 2T$ のとき, 波の先端は,

$$x = \underbrace{d}_{\text{点P}} + C \cdot 2T = 5d$$

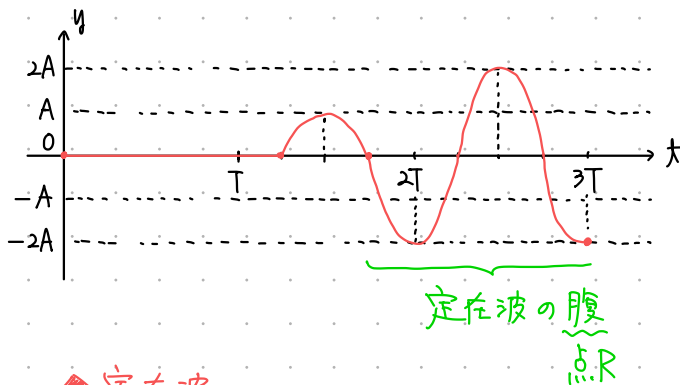


向4 点R に入射波が到達するのは,

$$\frac{3.5d - d}{C} = \frac{5}{4}T \text{ (後)}$$

点R に反射波が到達するのは,

$$\frac{5}{4}T + \frac{d}{C} = \frac{7}{4}T \text{ (後)}$$



向5. ◆ 定在波

* 反射による定在波

{ 自由端 : 必ず腹
固定端 : 必ず節

* 両側から λ, T, A が等しい波を重ね合わせるとき

{ 同位相 : 中央は腹
逆位相 : 中央は節

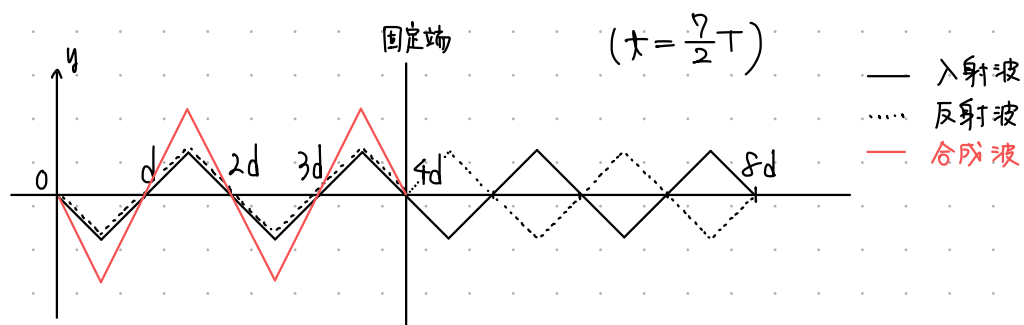
* 節(腹) は, $\frac{\lambda}{2}$ ごとにあり

節は $\frac{\lambda}{2} = d$ ごとにあり

$$x = 3d$$

問6 まず、反射波が $x=0$ に到達するのは、

$$\frac{3d+4d}{c} = \frac{7}{2}T \text{ (後)}$$



上図の $\frac{1}{4}T$ 後 $x=0 \sim 4d$ では $y=0$ 変位が全 0 .

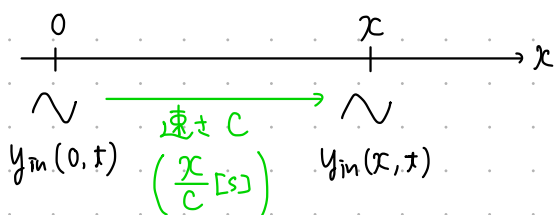
$$\frac{7}{2}T + \frac{1}{4}T = \frac{15}{4}T$$

$\frac{1}{4}T$ ↑

2

向1. $c = \frac{\frac{3}{4}\lambda}{\frac{1}{4}t_1} = \frac{3\lambda}{4t_1}$, $f = \frac{c}{\lambda} = \frac{3}{4t_1}$

向2. <入射波 y_{in} について>



$$y_{in}(0, t) = +a \sin(2\pi f t) = a \sin\left(2\pi \cdot \frac{3}{4t_1} t\right)$$

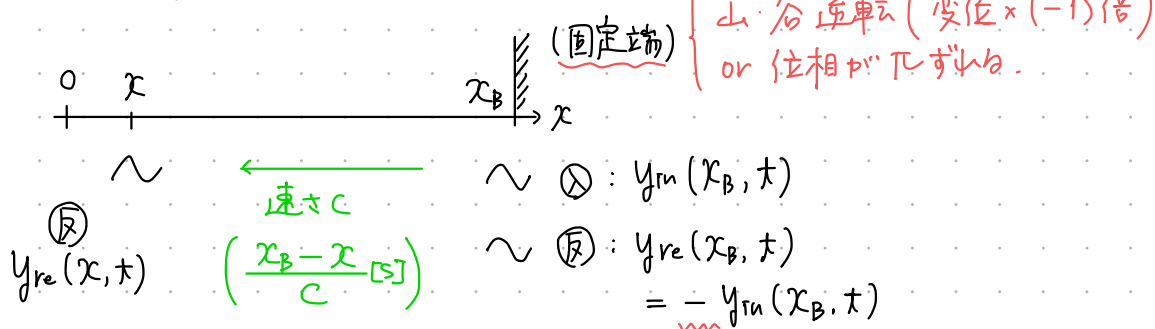
$$y_{in}(x, t) = y_{in}\left(0, t - \frac{x}{c}\right)$$

$x = x'$ の振動 $\frac{x}{c}$ [s] 前の原点の振動

$$= a \sin\left\{2\pi f\left(t - \frac{x}{c}\right)\right\}$$

$$= a \sin\left\{2\pi\left(ft - \frac{x}{\lambda}\right)\right\} \quad (\because c = f\lambda) \quad \dots ①$$

<反射波 y_{re} について>



$$y_{re}(x, t) = - y_{in}\left(x_B, t - \frac{x_B - x}{c}\right) \quad \left(= - y_{in}\left(0, \frac{2x_B - x}{c}\right) \right)$$

$x = x'$

$\frac{x_B - x}{c}$ [s] 前の $x = x_B$ の振動

$$= a \sin\left[2\pi\left\{f\left(t - \frac{x_B - x}{c}\right) - \frac{x_B}{\lambda}\right\}\right]$$

$$= a \sin\left\{2\pi\left(ft - \frac{2x_B - x}{\lambda}\right)\right\} \quad (\because c = f\lambda) \quad \dots ② //$$

波の先端が点Aに到達したとき ($t_A = \frac{x_A}{C}$), ①より

$$\begin{aligned} y_{in}(x, t_A) &= a \sin \left\{ 2\pi \left(f \cdot \frac{x_A}{C} - \frac{x}{\lambda} \right) \right\} \\ &= a \sin \left(2\pi \frac{x_A - x}{\lambda} \right) \quad (\because C = f\lambda) \end{aligned}$$

波(反射波)の先端が点Aに到達したとき ($t'_A = \frac{x_B + (x_B - x_A)}{C}$), ②より

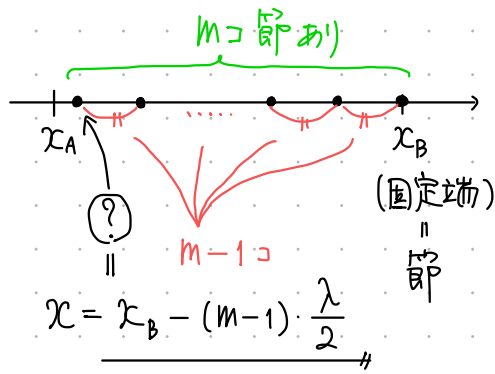
$$\begin{aligned} y_{re}(x, t'_A) &= -a \sin \left\{ 2\pi \left(f \cdot \frac{2x_B - x_A}{C} - \frac{x}{\lambda} \right) \right\} \\ &= -a \sin \left(2\pi \frac{x - x_A}{\lambda} \right) \quad (\because C = f\lambda) \\ &= a \sin \left(2\pi \frac{x_A - x}{\lambda} \right) \quad (\because \sin(-\theta) = -\sin\theta) \end{aligned}$$

$$\begin{aligned} y_{in}(x, t'_A) &= a \sin \left\{ 2\pi \left(f \cdot \frac{2x_B - x_A}{C} - \frac{x}{\lambda} \right) \right\} \quad (\because \text{①}) \\ &= a \sin \left(2\pi \frac{2x_B - x_A - x}{\lambda} \right) \quad (\because C = f\lambda) \end{aligned}$$

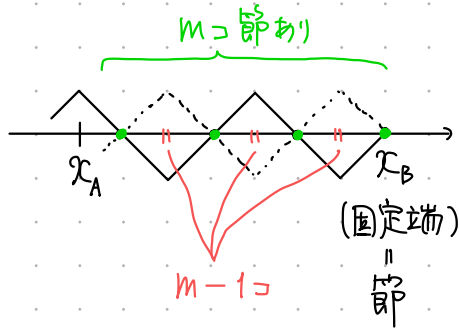
合成波は,

$$\begin{aligned} Y(x, t'_A) &= y_{in}(x, t'_A) + y_{re}(x, t'_A) \\ &= a \sin \left(2\pi \frac{x_A - x}{\lambda} \right) + a \sin \left(2\pi \frac{2x_B - x_A - x}{\lambda} \right) \\ &= 2a \sin \left(2\pi \frac{x_B - x}{\lambda} \right) \cos \left(2\pi \frac{x_B - x_A}{\lambda} \right) \\ &\quad \left(\because \sin \alpha + \sin \beta = 2 \sin \frac{\alpha + \beta}{2} \cos \frac{\alpha - \beta}{2} \right) \quad \left(\begin{array}{l} \uparrow: \frac{x_A - x_B}{\lambda} \\ \uparrow: \text{波長} \end{array} \right) \end{aligned}$$

問3



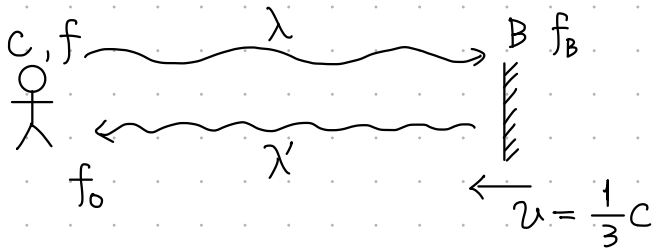
問4



$$x_B - x_A = (m-1) \frac{\lambda}{2} + \frac{\lambda}{4}$$

$$\therefore \lambda = \frac{4(x_B - x_A)}{2m-1}$$

問5



入射波の波長は, λ のまま.

B が観測するのは,

$$f_B = \frac{c + \frac{1}{3}c}{\lambda} = \frac{4}{3}f \quad (\because c = f\lambda)$$

B が出す波の波長は,

$$\lambda_B = \frac{c - \frac{1}{3}c}{f_B} = \frac{1}{2}\lambda$$

観測する振動数は,

$$f_0 = \frac{c}{\lambda_B} = 2 \cdot \frac{c}{\lambda} = 2f$$

3

問1 図より

$$\lambda_A = \frac{1}{2} OA = \underline{0.5 \text{ m}}, \quad \lambda_B = 2OB = \underline{1 \text{ m}}$$

問2 弦を伝わる波の速さは,

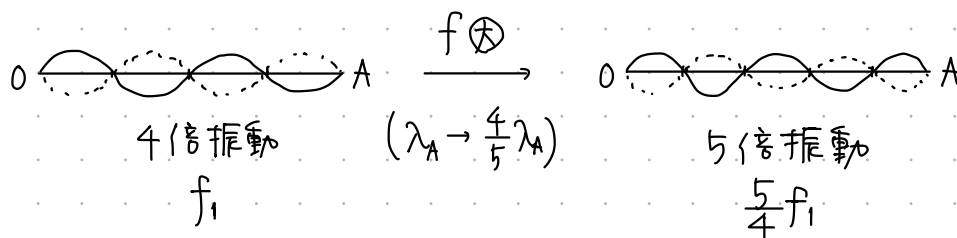
$$v_A = \sqrt{\frac{M_A g}{\rho}}, \quad v_B = \sqrt{\frac{M_B g}{\rho}} \quad \dots \textcircled{1}$$

これを代入すると,

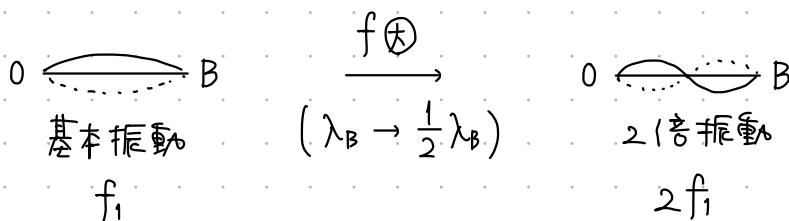
$$f_1 = \frac{v_A}{\lambda_A} = \frac{v_B}{\lambda_B}$$

$$\therefore \frac{v_B}{v_A} = \frac{\lambda_B}{\lambda_A} = 2 \quad \therefore \sqrt{\frac{M_B}{M_A}} = 2 \quad \therefore M_B = \underline{4M_A}$$

問3. <OA>



<OB>



→ Aの共振(共鳴)が先に起きる

$$f_2 = \underline{\underline{\frac{5}{4} f_1}}$$

問4. OAの共振振動数は,

$$f_1, \frac{5}{4}f_1, \frac{6}{4}f_1, \frac{7}{4}f_1, \underline{\frac{8}{4}f_1}, \dots$$

4倍 5倍 6倍

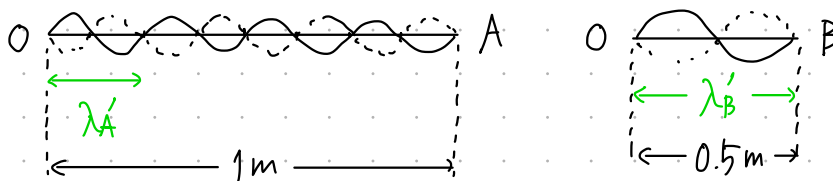
OBの共振振動数は,

$$f_1, \underline{2f_1}, 3f_1, \dots \quad \therefore f_3 = \underline{2f_1}$$

基本 2倍

このとき,

$$\lambda'_A = \frac{4}{8} \lambda_A = \underline{0.25 \text{ m}}, \quad \lambda'_B = \frac{1}{2} \lambda_B = \underline{0.5 \text{ m}}$$



問5. Aには6倍振動(振動数 $\frac{6}{4}f_1$)が生じているから,

$$f_1 + 220 = \frac{6}{4}f_1 \quad \therefore f_1 = \underline{440 \text{ Hz}}$$

$$\therefore v_A = \underbrace{f_1 \lambda_A}_{A \text{ が } 4 \text{ 倍振動のとき}} = \underline{220 \text{ m/s}}$$

Aが4倍振動のとき.

$$(6) \quad f_1 = \frac{v_B}{\lambda_B} = \frac{1}{\lambda_B} \sqrt{\frac{M_B g}{\rho}}, \quad f_1 + \Delta f = \frac{1}{\lambda_B} \sqrt{\frac{(M_B + \Delta M)g}{\rho}} \quad \text{より,}$$

$$\frac{f_1 + \Delta f}{f_1} = \sqrt{1 + \frac{\Delta M}{M_B}} \approx 1 + \frac{1}{2} \frac{\Delta M}{M_B} \quad \therefore \Delta f \approx \frac{\Delta M}{2M_B} f_1.$$

(7) $|f_1 - f_4| = 2 \text{ Hz}$ であるが, $\Delta f = \frac{1}{2} \times 0.04 \times 440 \text{ Hz} \approx 8.8 \text{ Hz}$ であることより, $f_1 < f_4$ と分かるので,

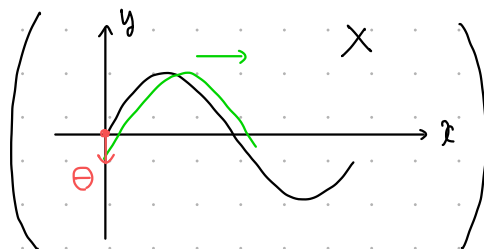
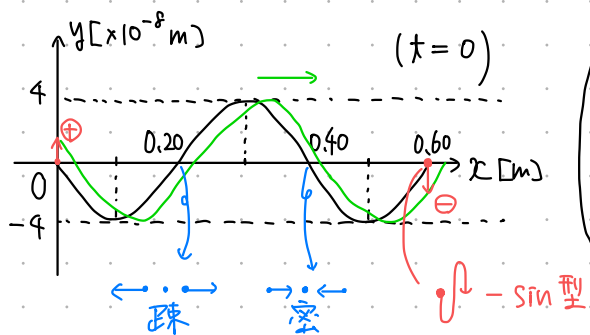
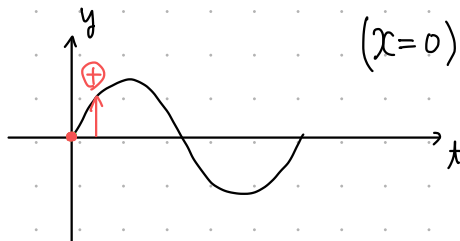
$$f_4 = \underline{442 \text{ Hz}}.$$

4

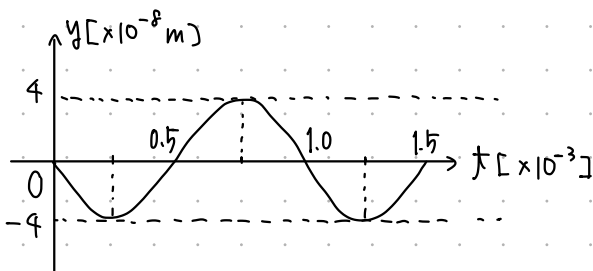
16) 1 図から,

(1) $T = 1.0 \times 10^{-3} \text{ s}$, $f = \frac{1}{T} = 1.0 \times 10^3 \text{ Hz}$, $\lambda = vT = 0.40 \text{ m}$
 $\omega = \frac{2\pi}{T} = 2.0\pi \times 10^3 \text{ rad/s}$

(2)

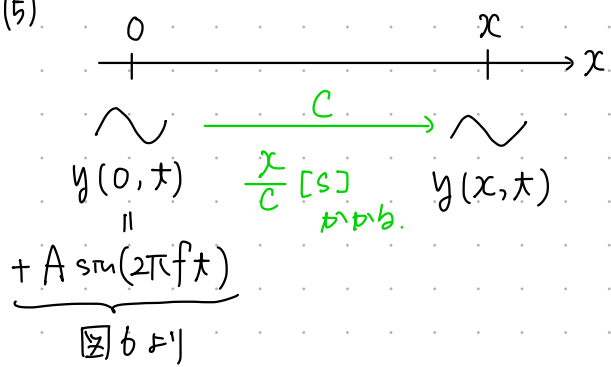


(3) $y(x=0.60 \text{ m}, t) = -A \sin\left(\frac{2\pi}{T}t\right)$



(4) $x = 0.2 \text{ m}, 0.6 \text{ m}$

(5)



$$\begin{aligned}
 y(x, t) &= y\left(0, t - \frac{x}{c}\right) \\
 &= A \sin\left(2\pi f\left(t - \frac{x}{c}\right)\right) \\
 &= A \sin\left(2\pi\left(ft - \frac{x}{\lambda}\right)\right) \quad (\because c = f\lambda)
 \end{aligned}$$

$$\begin{aligned}
 \therefore A &= \underline{4 \times 10^{-8} \text{ m}}, \quad B = f = \underline{1.0 \times 10^3 \text{ Hz}} \\
 C &= -\frac{1}{\lambda} = \underline{-2.5 \text{ m}^{-1}}
 \end{aligned}$$

$$\text{[6] 6. } y(x=0.5 \text{ m}, t=4.0 \times 10^{-3} \text{ s}) = \underline{-4 \times 10^{-8} \text{ m}}$$

II(1) 節と節の間隔は,

$$\frac{\lambda}{2} = \underline{0.20\text{ m}} .$$

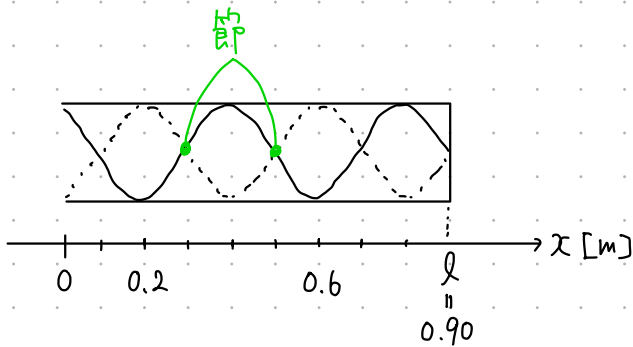
(2) 管の中に半波長が 4.5 個分生じるので (開口端補正は無視),

$$\ell = \frac{9\lambda}{4} = \underline{0.90\text{ m}} .$$

(3) ピストンの位置が節であり, そこから半波長ごとに節がある. よって, 範囲内にある節の位置は,

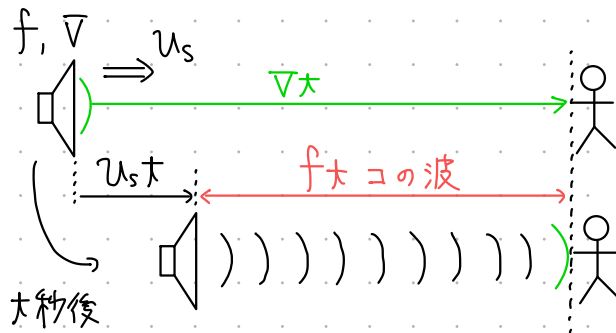
$$\frac{3\lambda}{2}, \frac{\lambda}{2} = \underline{0.30\text{ m}, 0.50\text{ m}} .$$

(1)



◆ ドップラー効果

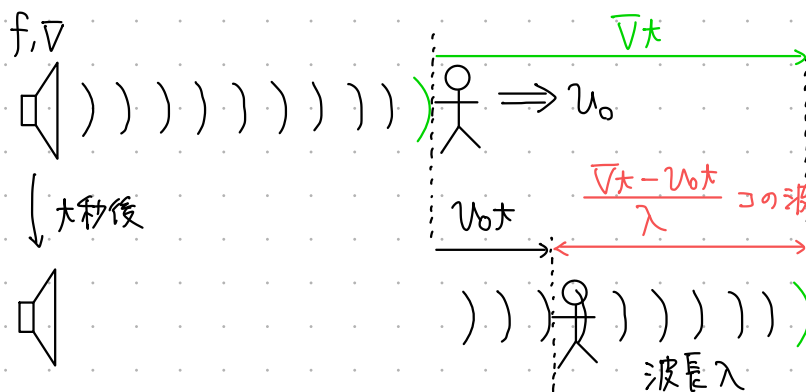
① 音源が動く場合 → 波の波長が変化する。



生じる波長は,

$$\lambda' = \frac{v \cdot t - u_s \cdot t}{f \cdot t} = \frac{v - u_s}{f}$$

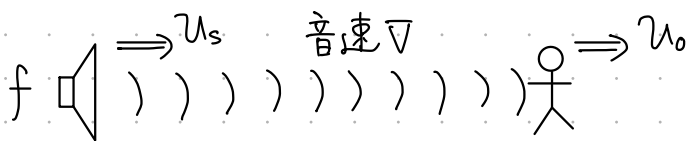
② 観測者が動く場合 → うけとる波の個数に変化する。



観測する振動数は,

$$f' = \frac{v \cdot t - u_o \cdot t}{\lambda} / t = \frac{v - u_o}{\lambda}$$

<まとめ>



* 生じる波長: $\lambda' = \frac{v - u_s}{f}$ (音源から見た音速)

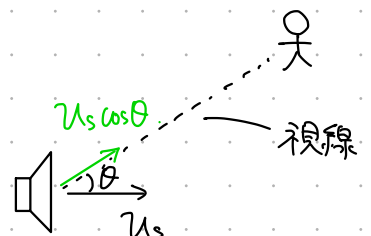
* 観測する振動数: $f' = \frac{v - u_o}{\lambda'}$ (観測者から見た音速)

$$= \frac{v - u_o}{v - u_s} f$$

(風がある場合は, $v \rightarrow v + w$ だとする)

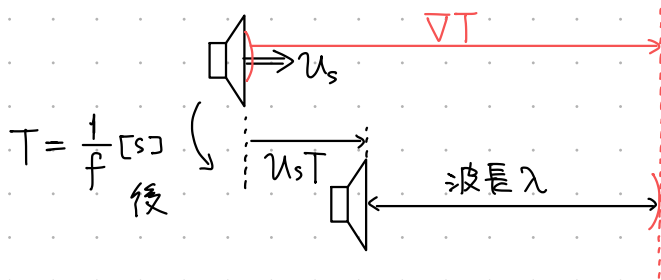
風速 w の風が音と同じ向きにあるとき.

(音源と観測者の運動方向が一直線上にない場合は,
 音源・観測者の速度の視線成分のみ寄与)



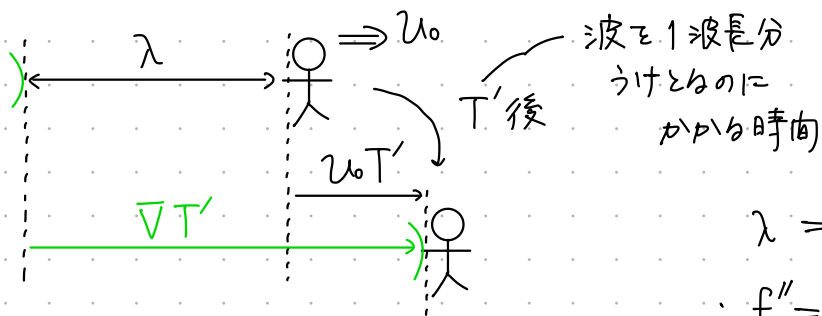
◆ 公式導出② ～ 1周期分の波に注目～

- (1) 音源が動く場合 → 波長が変わる.
振動数を f , 音速を v とする.



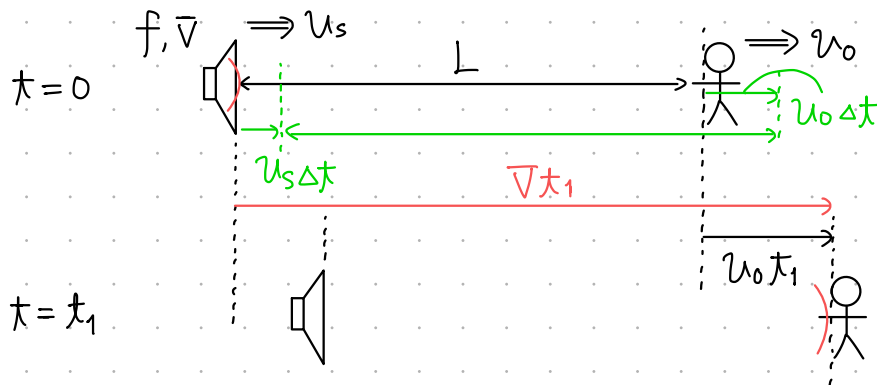
$$\begin{aligned}\lambda' &= (v - u_s)T \\ &= \frac{v - u_s}{f} \\ \therefore f' &= \frac{v}{\lambda'} = \frac{v}{v - u_s} f\end{aligned}$$

- (2) 観測者が動く場合 → うけとる波の個数が変わる.



$$\begin{aligned}\lambda &= (v - u_o)T' \\ \therefore f'' &= \frac{1}{T'} = \frac{v - u_o}{\lambda} \\ &= \frac{v - u_o}{v} f\end{aligned}$$

◆ 公式導出 ③ ～ 時間差に注目 ～



$t=0$ に、音源で出た波面が $t=t_1$ に観測者に届く

$$\underline{L} + \underline{u_o t_1} = \underline{V t_1} \quad \dots ①$$

$t=\Delta t$ に音源で出た波面が $t=t_1 + \Delta t$ に観測者に届く

$$\underline{L + (u_o - u_s) \Delta t} + \underline{u_o (t_1 + \Delta t - \Delta t)} = \underline{V (t_1 + \Delta t - \Delta t)} \quad \dots ②$$

② - ① より

$$u_o \Delta t_1 - u_s \Delta t = V (\Delta t_1 - \Delta t)$$

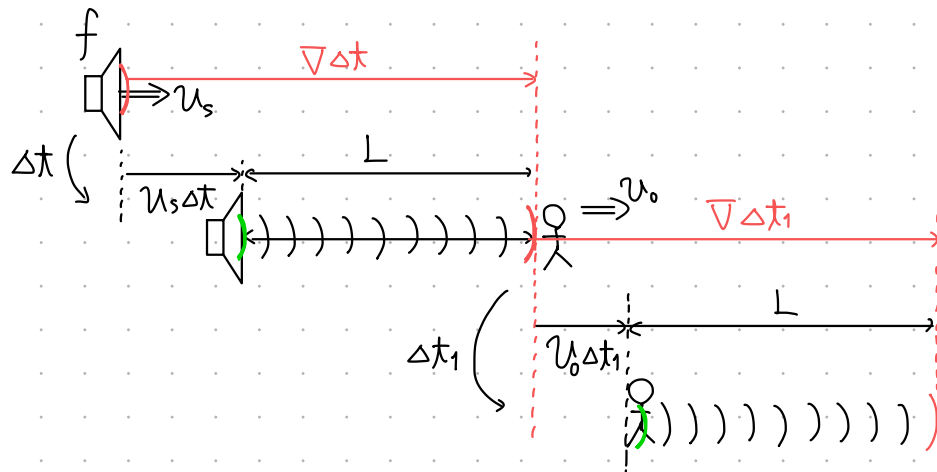
$$\therefore \Delta t_1 = \frac{V - u_s}{V - u_o} \Delta t \quad \dots ③$$

$$\left(\underline{(V - u_s) \Delta t} = \underline{(V - u_o) \Delta t_1} \right) \leftarrow \begin{array}{l} \text{音源から} \Delta t \text{ 間音を出して、} \\ \text{観測者が} \Delta t_1 \text{ 間} \\ \text{音を観測したとき} \end{array}$$

音源視点での
波の全長

観測者視点での
波の全長

もしくは、...



$$L = (v - u_s) \Delta t = (v - u_o) \Delta t_1$$

$$\therefore \Delta t_1 = \frac{v - u_s}{v - u_o} \Delta t \quad \dots (3)$$

観測者は、 $f \Delta t$ の波を Δt_1 だけ受けとるので、(3)より

$$f' = \frac{f \Delta t}{\Delta t_1} = \frac{v - u_o}{v - u_s} f$$

5

I $t = 0$ に S を出た波面が, $t = t_1$ に O に届くので,

$$t_1 = \frac{x_0}{\underbrace{V}_{(1)}} .$$

$t = 0 \sim t_1$ の間に S は O へ vt_1 だけ近づきつつ, $\underbrace{f_0 t_1}_{n}$ 個の波を出すので,

$$\left(\lambda = \frac{x_0 - vt_1}{f_0 t_1} = \frac{V - v}{f_0} \right) . \quad \underbrace{\frac{f_0 x_0}{V}}_{(2)}$$

$t = t_1$ に S を出た波面が, $t = t_2$ に O に届くので,

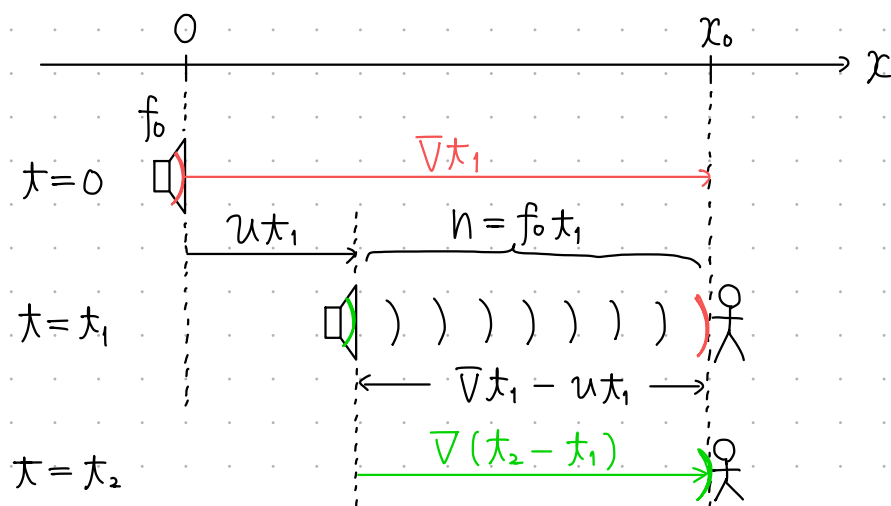
$$\frac{x_0 - vt_1}{\#} = V(t_2 - t_1) \quad \therefore t_2 = \frac{x_0(2V - v)}{\underbrace{V^2}_{(3)}} . \quad \leftarrow \begin{array}{l} \text{波の端} \sim \text{端までの} \\ \text{長さを2通りで表す} \end{array}$$

もしくは, $Vt_1 - vt_1$

すると, O は時間 $t_2 - t_1$ の間に $\underbrace{f_0 t_1}_n$ 個の波を受けとる. よって, 観測する振動数 (単位時間あたりに受ける波の個数) は^{*1},

$$f_1 = \frac{f_0 t_1}{t_2 - t_1} = \frac{V}{\underbrace{V - v}_{(4)}} f_0 . \quad \leftarrow \underbrace{f_1(t_2 - t_1)}_{\text{観測者がうける波の数}} = \underbrace{f_0 t_1}_{\text{音源が出す波の数}}$$

$$\left(\begin{array}{l} \text{さらに O が動いている場合には}^{*2}, \\ f_1^* = \frac{V - u}{\lambda} = \frac{V - u}{V - v} f_0 \end{array} \right)$$



II S → O と伝わる音波の音速が $\underbrace{V+U}_{(1)}$ になると考えればよい^{※3}. t_1, t_2, f_1 の式中の V を $V+U$ で置き換えて^{※4},

$$t_3 = \frac{x_0}{\underbrace{V+U}_{(2)}}, \quad t_4 = \frac{x_0 \{2(V+U) - v\}}{\underbrace{(V+U)^2}_{(3)}}, \quad f_2 = \frac{V+U}{\underbrace{V+U-v}_{(4)}} f_0.$$

III 作図により,

$$x_f = \sqrt{(Vt_f)^2 - (Ut_f)^2} = \underbrace{\sqrt{V^2 - U^2}t_f}_{(1)}.$$

よって, x 軸に沿って音が伝わる速さは,

$$V_x = \frac{x_f}{t_f} = \underbrace{\sqrt{V^2 - U^2}}_{(2)}.$$

すると, f_1 の $\overline{V} \neq \overline{V}_x = \sqrt{\overline{V}^2 - U^2}$ にかきかえて,

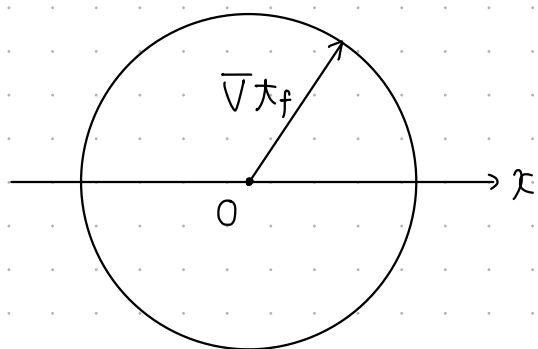
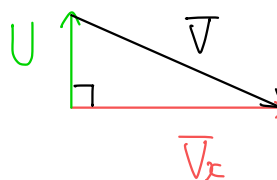
$$f_3 = \frac{V_x}{V_x - v} f_0 = \frac{\sqrt{V^2 - U^2}}{\underbrace{\sqrt{V^2 - U^2} - v}_{(3)}} f_0.$$

さて,

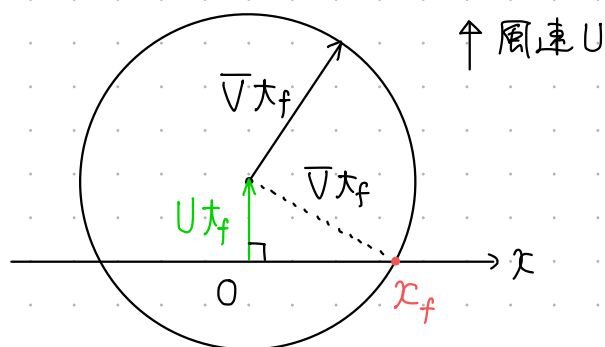
$$\frac{f_3 - f_0}{f_0} = \frac{v}{\sqrt{V^2 - U^2} - v} = \frac{\frac{v}{V}}{\sqrt{1 - \left(\frac{U}{V}\right)^2} - \frac{v}{V}} = \frac{\frac{v}{V}}{1 - \frac{1}{2}\left(\frac{U}{V}\right)^2 - \frac{v}{V}}.$$

$\frac{U}{V} = 25\% = \frac{1}{4}, \quad \frac{v}{V} = 10\% = \frac{1}{10}$ のとき,

$$\frac{f_3 - f_0}{f_0} = \frac{16}{139} = 0.115 \dots \approx \underbrace{12\%}_{(4)}.$$



(風みし)



(風みし)

6

(1) 略.
$$f = \frac{\bar{V}}{\bar{V} - u_s} f_0$$

(2) 各点で発せられた音波が点 P で観測される際の振動数は、それぞれ、

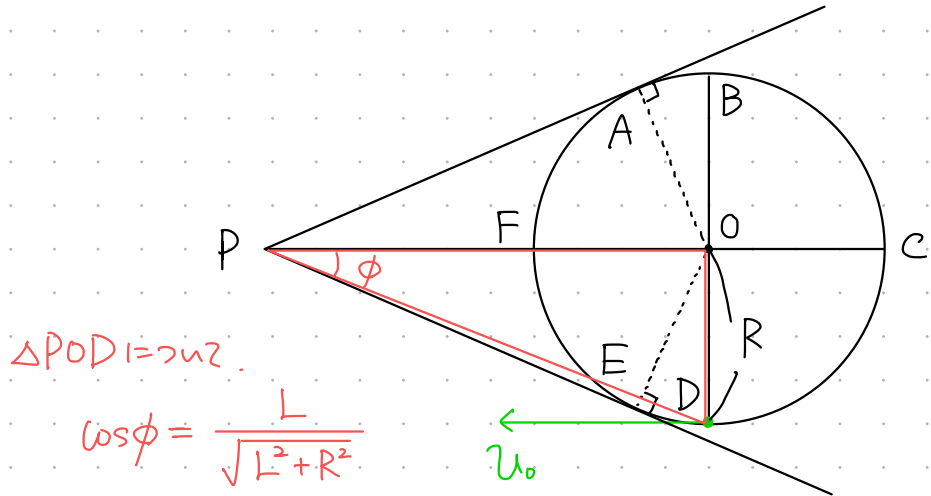
$$f_A = \frac{V}{\underbrace{V + v_0}} f_0, \quad f_B = \frac{V}{\underbrace{V + v_0 \frac{L}{\sqrt{L^2 + R^2}}}} f_0, \quad f_C = \underbrace{f_0},$$

$$\underbrace{f_D = \frac{V}{V - v_0} \frac{L}{\sqrt{L^2 + R^2}} f_0}_{\text{Doppler shift}}, \quad \underbrace{f_E = \frac{V}{V - v_0} f_0}_{\text{Doppler shift}} = f_{\text{max}}, \quad f_F = \underbrace{f_0}_{\text{rest frequency}}.$$

$$f_{\max} : \underline{\mathbb{E}} \qquad f_{\min} : \underline{\mathbb{A}}$$

$$(3) \quad f_{\min} = \frac{V}{V + v_0} f_0, \quad f_{\max} = \frac{V}{V - v_0} f_0 \text{ ㄹ ㄱ,}$$

$$v_0 = \frac{f_{\max} - f_{\min}}{f_{\max} + f_{\min}} V, \quad f_0 = \frac{2f_{\max}f_{\min}}{f_{\max} + f_{\min}}.$$



$$(3) \quad \begin{cases} f_{\min} = \frac{V}{V + u_0} f_0 & \dots \textcircled{1} \\ f_{\max} = \frac{V}{V - u_0} f_0 & \dots \textcircled{2} \end{cases}$$

① / ② よい

$$\frac{f_{\min}}{f_{\max}} = \frac{V - u_0}{V + u_0} \quad \therefore u_0 = \frac{f_{\max} - f_{\min}}{f_{\max} + f_{\min}} V$$

$$(4) \quad T = t_3 - t_1 = \frac{2\pi R}{v_0} \text{ より,}$$

$$R = \frac{v_0(t_3 - t_1)}{2\pi}.$$

$$(5) \quad t_3 - t_1 = \frac{2\pi}{\omega}, \quad t_3 - t_2 = \frac{\pi - 2\theta}{\omega} \text{ より, 辺々を割ると, } \frac{t_3 - t_2}{t_3 - t_1} = \frac{\pi - 2\theta}{2\pi}$$

音源の
角速度

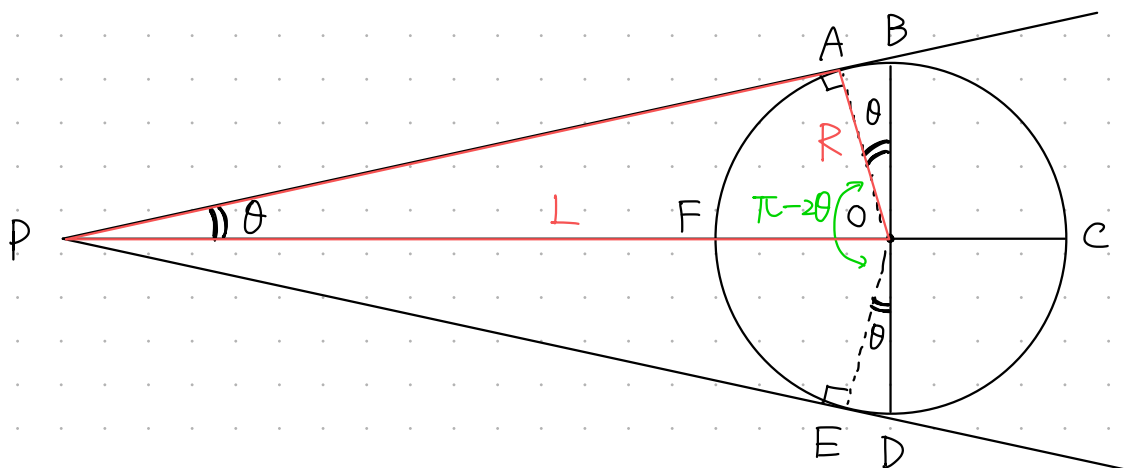
$$\theta = \frac{\pi}{2} - \frac{t_3 - t_2}{t_3 - t_1} \pi.$$

すると,

$$R = \frac{v_0(t_3 - t_1)}{2\pi} = \frac{f_{\max} - f_{\min}}{f_{\max} + f_{\min}} \frac{V(t_3 - t_1)}{2\pi},$$

$$L = \frac{R}{\sin \theta} = \frac{f_{\max} - f_{\min}}{f_{\max} + f_{\min}} \frac{V(t_3 - t_1)}{2\pi \cos \left(\frac{t_3 - t_2}{t_3 - t_1} \pi \right)}.$$

$$\sin \left(\frac{2t_2 - t_1 - t_3}{2(t_3 - t_1)} \pi \right) = \sin \left(\frac{\pi}{2} - \frac{t_3 - t_2}{t_3 - t_1} \pi \right)$$



7

(1) 屈折の法則より,

$$1 \cdot \sin \theta_0 = n_1 \sin \theta_1 \quad \therefore \sin \theta_1 = \frac{\sin \theta_0}{\underbrace{n_1}}$$

(2) $n_1 \sin(90^\circ - \theta_1) = n_2 \sin \phi$ より,

$$\sin \phi = \frac{n_1}{n_2} \cos \theta_1 = \frac{n_1}{n_2} \sqrt{1 - \sin^2 \theta_1} = \frac{1}{n_2} \sqrt{n_1^2 - \sin^2 \theta_0}$$

全反射が起こる条件 (屈折角 ϕ が存在しない条件) は, $\sin \phi > 1$ として,

$$\sqrt{n_1^2 - \sin^2 \theta_0} > n_2 \quad \therefore \sin \theta_0 < \underbrace{\sqrt{n_1^2 - n_2^2}}.$$

(3) 図略.

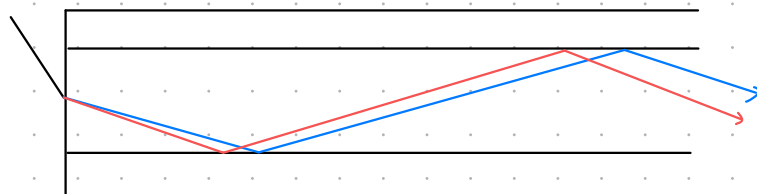
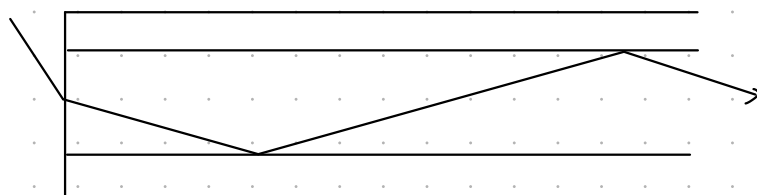
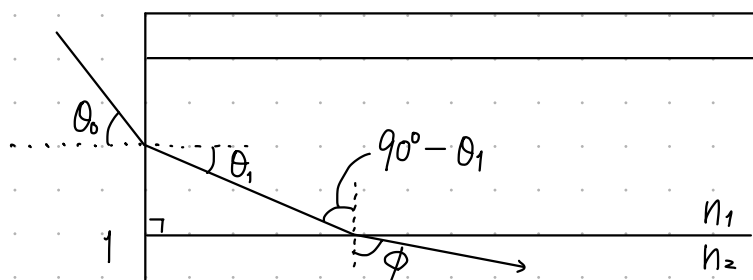
白色光は, 様々な波長の光を含む. また, 光の分散現象により, 波長が長い光ほど屈折率がわずかに小さく, 屈折角 θ_1 が大きくなる. そのため経路はスペクトル状に分かれる. (78 字)

(4) 角 α を次のように定める:

$$\tan \alpha = \frac{d/2}{F}, \quad \sin \alpha = \frac{d/2}{\sqrt{F^2 + (d/2)^2}}.$$

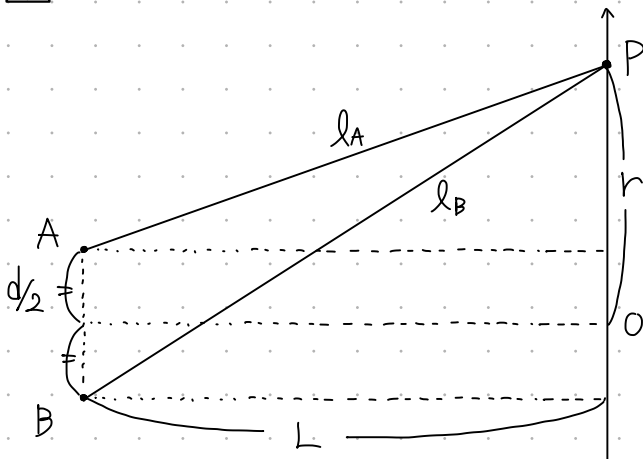
$\theta_0 = \alpha$ が全反射の条件を満たせば, 全ての光が全反射する. そのためには, $\sin \alpha < \sqrt{n_1^2 - n_2^2}$ より,

$$\frac{d/2}{\sqrt{F^2 + (d/2)^2}} < \sqrt{n_1^2 - n_2^2} \quad \therefore F > \underbrace{\frac{\sqrt{(1 - n_1^2 + n_2^2)d/2}}{\sqrt{n_1^2 - n_2^2}}} (= f).$$





9



◆ 道のり $L \longleftrightarrow$ 位相 $\frac{2\pi}{\lambda} L$

問1

(1) 強め合いの条件は,

$$\underbrace{\frac{2\pi}{\lambda} (l_B - l_A)}_{\text{位相差}} = 2n\pi \quad (n = 0, 1, 2, \dots)$$

$$\therefore \underbrace{l_B - l_A}_{\text{経路差}} = n\lambda$$

$$(2) \quad l_A = \sqrt{L^2 + \left(r - \frac{d}{2}\right)^2}, \quad l_B = \sqrt{L^2 + \left(r + \frac{d}{2}\right)^2}$$

$$(3) \quad l_A = L \left\{ 1 + \underbrace{\left(\frac{r - d/2}{L} \right)^2}_{\ll 1} \right\}^{\frac{1}{2}}$$

$$\doteq L \left\{ 1 + \frac{1}{2} \left(\frac{r - d/2}{L} \right)^2 \right\} = \underline{L + \frac{1}{2L} \left(r - \frac{d}{2} \right)^2}$$

$$l_B = \underline{L + \frac{1}{2L} \left(r + \frac{d}{2} \right)^2}$$

(4) (3) より

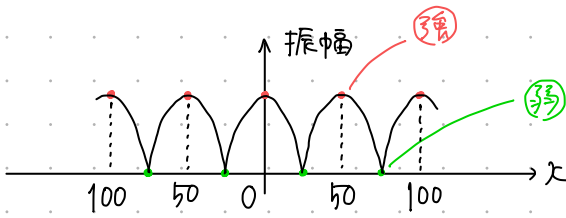
$$l_B - l_A = \frac{dr}{L} = n\lambda \quad (\because (1))$$

$$\therefore r = \underline{n \frac{L\lambda}{d}}$$

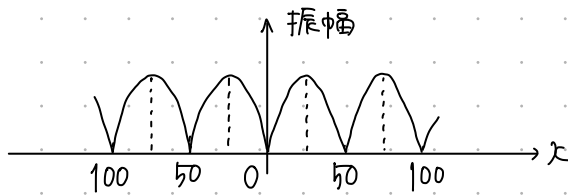
問2.

(1) 強め合う位置は,

$$r = n \frac{L\lambda}{d} = 50n \text{ [cm]}$$

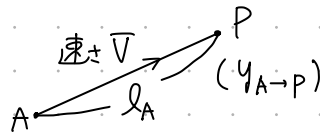


(2) A, B の位相



(3) 波源 A の振動の式を.

$$y_A = a \sin\left(\frac{2\pi}{T} t\right) \quad (y_A)$$



とすると, 点 P での振動は,

$$\begin{aligned} y_{A \rightarrow P} &= a \sin\left\{ \frac{2\pi}{T} \left(t - \frac{l_A}{v} \right) \right\} \quad (v: \text{波の伝わる速度}) \\ &= a \sin\left(\frac{2\pi}{T} t - \underbrace{\frac{2\pi}{T} \frac{l_A}{v}}_{\frac{l_A}{\lambda} \text{ [s] 前}} \right) \quad (\lambda = vT) \\ &= a \sin\left(\frac{2\pi}{T} t - \frac{2\pi}{\lambda} l_A \right) \quad (\theta_A) \end{aligned}$$

波源 B の振動の式は,

$$y_B = a \sin\left(\frac{2\pi}{T} t + \phi\right)$$

とすれば, 点 P での振動は,

$$\begin{aligned} y_{B \rightarrow P} &= a \sin\left\{ \frac{2\pi}{T} \left(t - \frac{l_B}{v} \right) + \phi \right\} \\ &= a \sin\left(\frac{2\pi}{T} t - \underbrace{\frac{2\pi}{T} \frac{l_B}{v}}_{\frac{l_B}{\lambda} \text{ [s] 前}} + \phi \right) \\ &= a \sin\left(\frac{2\pi}{T} t - \frac{2\pi}{\lambda} l_B + \phi \right) \quad (\theta_B) \end{aligned}$$

よ、2, 強め合いの条件は,

$$-\frac{2\pi}{\lambda} l_A - \left(-\frac{2\pi}{\lambda} l_B + \phi \right) = 2n\pi$$

$$\frac{2\pi}{\lambda} (l_B - l_A) - \phi = 2n\pi$$

$$\frac{2\pi}{\lambda} \frac{dr}{L} - \phi = 2n\pi$$

$$\therefore r = \frac{L\lambda}{d} \cdot \frac{\phi}{2\pi} + n \cdot \frac{L\lambda}{d}$$

$n=0$ のとき,

$$r = \frac{L\lambda}{d} \cdot \frac{\phi}{2\pi} = \underline{\underline{\frac{\phi}{2\pi} \times 70 [\text{cm}]}}$$

<補足>

点Pでの合成波は,

$$Y = y_{A \rightarrow P} + y_{B \rightarrow P}$$

$$= a \sin \theta_A + a \sin \theta_B$$

$$= 2a \sin \left(\frac{\theta_A + \theta_B}{2} \right) \cos \left(\frac{\theta_A - \theta_B}{2} \right)$$

$$= 2a \cos \left[\frac{1}{2} \left\{ \frac{2\pi}{\lambda} (l_B - l_A) - \phi \right\} \right] \\ \times \sin \left(\frac{2\pi}{T} \otimes - \frac{2\pi}{\lambda} \frac{l_A + l_B}{2} + \frac{\phi}{2} \right)$$

$$\longrightarrow \text{振幅} : 2a \left| \cos \left[\frac{1}{2} \left\{ \frac{2\pi}{\lambda} (l_B - l_A) - \phi \right\} \right] \right|$$

$$\text{強め合い条件} : \frac{1}{2} \left\{ \frac{2\pi}{\lambda} (l_B - l_A) - \phi \right\} = n\pi$$

$$\therefore \frac{2\pi}{\lambda} (l_B - l_A) - \phi = 2n\pi$$